

Requirements Specification (Pflichtenheft)

AI Researcher Radar

Lennart Specht

Moritz Stürmer

Gvantsa Tchuradze

Supervisor: Prof. Dr. Ingo Scholtes
May 9, 2026

1 Project Overview

The project **AI Researcher Radar** is a web-based platform for exploring and visualizing researchers in Artificial Intelligence and Computer Science. It integrates bibliographic data from public sources and presents it using interactive visualizations such as a world map and a collaboration network graph. The system supports research exploration, academic networking, and university talent scouting.

2 Goals

2.1 Must-Have Requirements

- Web application for searching and visualizing AI researchers
- Interactive world map of researchers
- Visualization of co-authorship and collaboration networks
- Researcher profiles with name, institution, research areas, publications, and citation metrics
- Search and filtering functionality
- Local database combining OpenAlex and DBLP data
- REST backend for data processing
- Web frontend with interactive visualization
- Clear separation of frontend and backend
- Maintainable and extensible architecture
- Dataset of approximately 500–1000 researchers

2.2 Midterm Goals

- Functional data retrieval from OpenAlex API
- Initial dataset with at least 200 researchers
- Basic GUI and navigation working
- Initial researcher profiles available

- Basic map or graph visualization implemented
- At least 50% of must-have requirements completed

2.3 Responsibilities

Frontend development (Gvantsa Tchuradze) focuses on implementing the user interface, including the interactive map, graph visualization, and search functionality. Backend development (Lennart Specht) is responsible for building the REST API, database design, and data integration services. Data engineering (Moritz Stürmer) handles data collection from OpenAlex and DBLP, preprocessing, deduplication, and preparation of datasets for use in the system.

3 Optional Requirements

- Ranking by citations, publication count, or h-index
- Advanced collaboration graph visualization
- Filtering by research area (e.g., NLP, Computer Vision)
- Filtering by geographic region
- Visualization of publication venues
- Highlighting highly connected researchers
- Heuristic gender estimation
- Improved UI/UX
- Enhanced deduplication methods

4 Exclusions

- No full coverage of all global AI researchers
- No real-time data updates
- No login or authentication system
- No mobile application
- No full DBLP dataset integration

5 Application and Target Users

5.1 Application Areas

- University talent scouting and recruiting
- Exploration of scientific collaboration networks
- Orientation for students and researchers

5.2 Target Group

- Professors and research staff
- Recruiters and institutions
- Computer Science and AI students

6 System Environment

6.1 Software

- GitLab for version control
- OpenAlex API and DBLP as data sources
- PostgreSQL database
- Python (FastAPI / Flask) backend
- React + TypeScript frontend
- Leaflet or Mapbox for map visualization
- Cytoscape.js (optional) for graph visualization

6.2 Hardware

- University-hosted virtual machine
- Internet connection required
- Modern web browser (Chrome, Firefox, Edge)

7 System Functionality

Users can explore AI researchers via an interactive map or graph. They can zoom, pan, search, and click on nodes to view detailed researcher profiles including publications, citations, affiliations, and collaborations. The backend continuously retrieves and processes data from OpenAlex and DBLP and stores it locally.

8 Data Management

The system stores researchers, institutions, publications, collaboration relationships, and citation metrics. Data is collected from OpenAlex and DBLP, merged locally, and duplicate entries are removed where possible.

9 Performance Requirements

Search results should be returned within a few seconds. Map and graph visualizations must load efficiently, and user interactions should remain responsive. Caching strategies may be used if necessary.

10 User Interface

The system includes an interactive world map or graph, a search bar with filters, a researcher detail panel, and optional advanced filtering tools.

11 Quality Goals

The system will be modular, well-structured, and maintainable. Frontend and backend are clearly separated. Comprehensive documentation will be provided, including a developer handbook and a user manual. The project emphasizes extensibility, usability, and documentation of AI-assisted development.